Hands-On Generative AI

Parameter-Efficient Fine-Tuning and Sampling

Day 3 - Session 2

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What We'll Cover

1. Why full fine-tuning is expensive
2. LoRA: train a tiny adapter, freeze the base
3. Demo: LoRA parameter math
4. Lab 3.2: LoRA on a small model
5. QLoRA and quantization (concept)
6. Sampling: temperature, top-p, top-k, beam
7. Demo: Sampling Knobs
8. Lab 3.3: sampling with generate()

Parameter-Efficient Fine-Tuning

Train a little, freeze the rest

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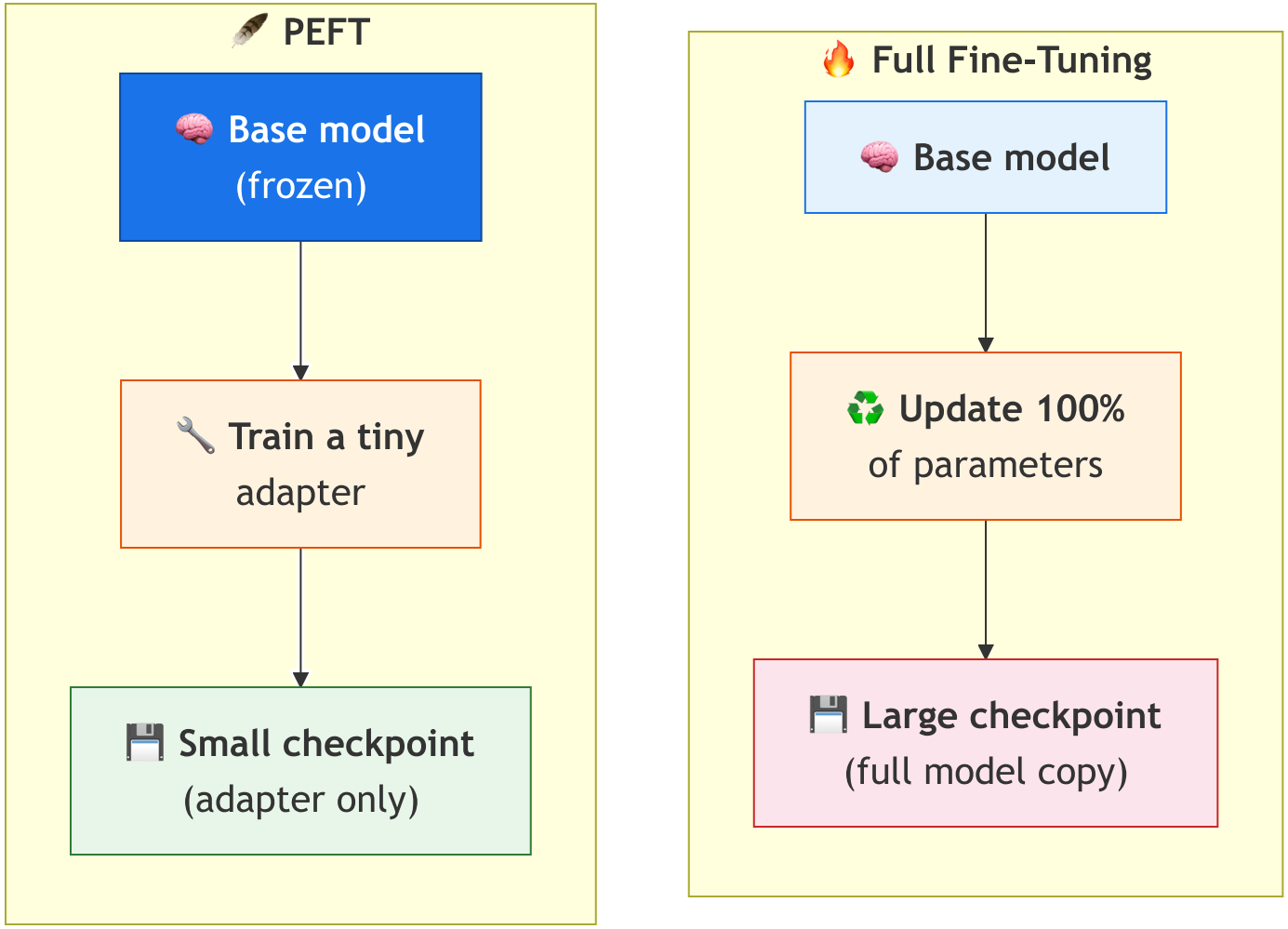
The Problem With Full Fine-Tuning

Updating every weight is expensive and produces a full-size checkpoint per task.

- Gradients and optimizer state multiply the memory cost
- Each task means storing another copy of the whole model
- The base model can drift and forget general ability

PEFT freezes the base and trains a small set of new parameters instead.

Full Fine-Tuning vs PEFT



What PEFT Buys You

Freeze the base, train a small adapter, and the numbers change fast.

- Far less GPU memory for gradients and optimizer state
- Small checkpoints in megabytes, swappable per task
- The frozen base keeps its general ability

> **IN PRACTICE**
>
> Many managed AI platforms only support fine-tuning via LoRA. They freeze their giant proprietary base model and give you a private adapter.

LoRA

Low-rank adapters

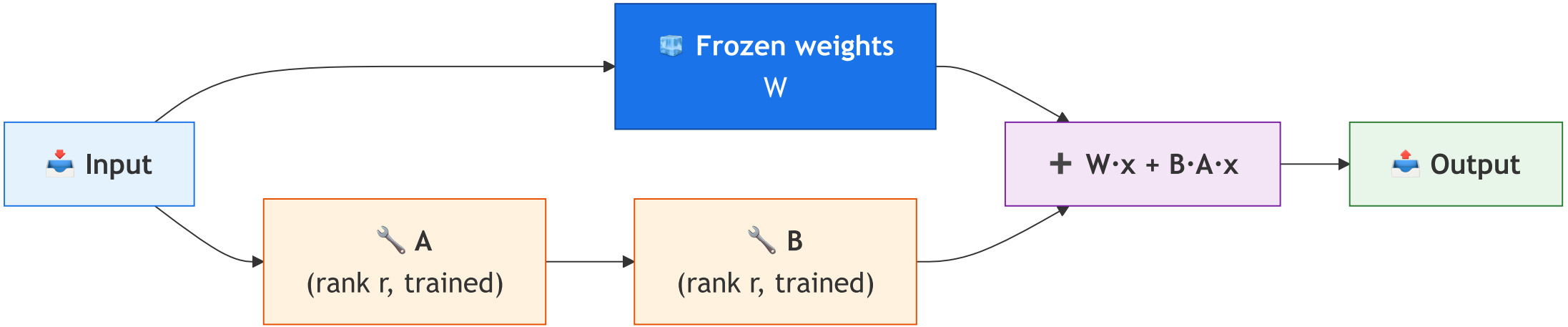
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The LoRA Idea

LoRA freezes the original weight matrix `W` and learns a small update as the product of two low-rank matrices `B` and `A`.

- Only `A` and `B` are trained; `W` stays frozen
- The rank `r` is small, so there are few new parameters
- At inference you add `B x A` back onto `W`

LoRA in One Picture



LoRA With the peft Library

```
from peft import LoraConfig, get_peft_model

config = LoraConfig(r=8, lora_alpha=16,
                    target_modules=["q_proj", "v_proj"],
                    lora_dropout=0.05, task_type="CAUSAL_LM")
model = get_peft_model(base_model, config)
model.print_trainable_parameters()  # often < 1% of total
```

Start with a small rank and raise it only if the model underfits.

> **TIP**
>
> Start with a low rank like r=8 or r=16. A large rank rarely improves performance and defeats the point of PEFT.

Demo: LoRA Parameter Math

Run `day3/demos/demo-lora-parameters.py`.

- Count params: a 4096 x 4096 matrix vs a rank-8 B x A adapter (< 0.4%)
- Factor a weight update with SVD and vary the rank
- See reconstruction error shrink as rank grows

> Low rank captures most of a weight change for few parameters.

Intuit Scenario: Mailchimp Brand Voices

PEFT allows serving hundreds of specialized models from one base.

- **Base model:** Instruct-tuned Llama 3 8B
- **Task:** Draft marketing emails in a specific customer's brand voice
- **Mechanism:** Train one tiny LoRA adapter per customer
- **Outcome:** The large base model stays resident in GPU memory while tiny customer adapters are swapped in and out per request

Breakout Lab 3.2

LoRA on a small model

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Lab 3.2: Train a LoRA Adapter

Open `day3/lab3_2_lora/start.ipynb` in the companion repo.

- Load `Qwen/Qwen2.5-0.5B-Instruct` (about 950 MB per student)
- Fallback: `SmolLM2-360M-Instruct` for weak wifi
- Attach a LoRA config and run a tiny training loop
- Compare generation before and after

> On slow wifi, use the smaller fallback model to keep the download short.

QLoRA and Quantization

Concept only

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QLoRA in Concept

Quantization stores weights with fewer bits, which shrinks memory use. QLoRA (quantized LoRA) loads the frozen base in 4-bit and trains LoRA adapters on top.

- Fine-tune a large model on a single smaller GPU
- Base stays in 4-bit; the small adapter trains in higher precision
- The 4-bit path needs a CUDA GPU (NVIDIA)

The labs use plain LoRA that runs on CPU or Apple Silicon.

Sampling

How a model turns probabilities into text

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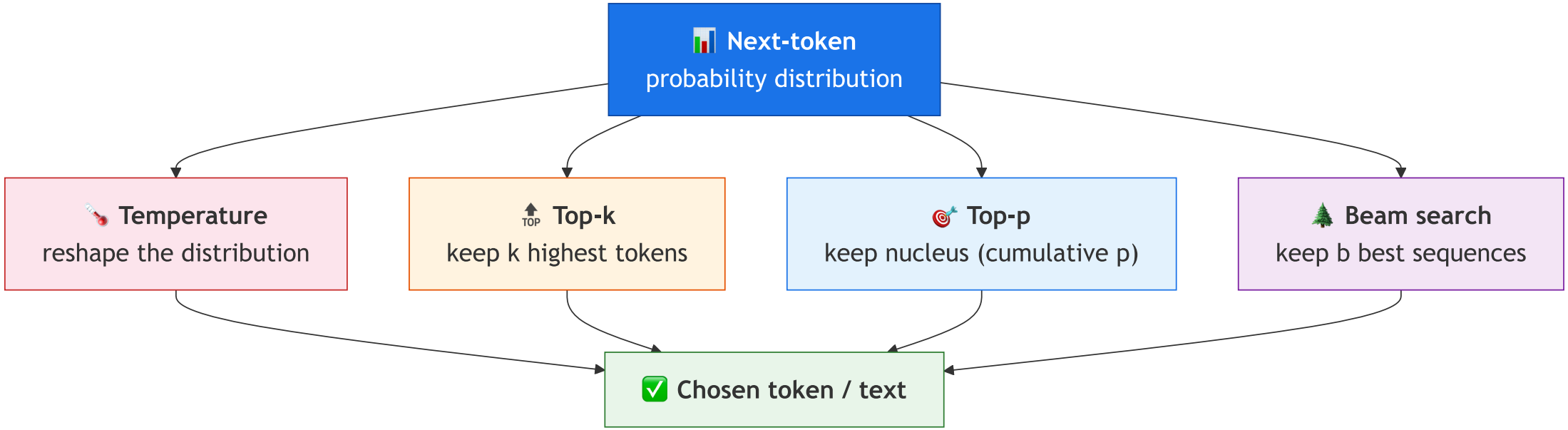
Greedy vs Sampling

At each step the model produces a probability over the whole vocabulary. Decoding picks the next token from it.

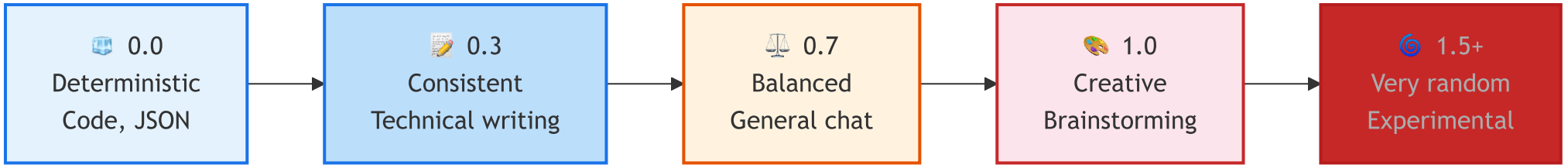
- **Greedy:** always take the top token. Deterministic, can repeat
- **Sampling:** draw a token according to the probabilities

The knobs below reshape the distribution before the draw.

The Sampling Knobs



Temperature: Visual Intuition



Temperature controls the **probability distribution**:

- **0.0:** Always picks the most likely word
- **0.7:** Balanced between predictable and creative
- **1.0+:** More random, creative, but potentially incoherent

What Each Knob Does

Knob	Effect
Temperature	Sharpen (low) or flatten (high) the distribution
Top-k	Keep the k highest-probability tokens
Top-p	Keep the nucleus reaching cumulative p
Beam search	Keep the b best sequences, pick the best

WARNING

Do not set both `temperature` and `top_p` to non-default values simultaneously unless you have a highly specific reason. They conflict in how they reshape the distribution.

Choosing Settings by Task

Task	temperature	top_p
Code, JSON	0.0	1.0
Summarization	0.3	1.0
General chat	0.7	1.0
Brainstorming	1.0	0.9

Beam search (num_beams 4) suits translation and summarization.

Demo: Sampling Knobs in Text Generation

Run `day3/demos/demo-sampling-knobs.py`.

- Run a small GPT-2 model on a single prompt
- Observe greedy output vs temperature variants
- See beam search select for high fluency

> Same prompt, different knobs, entirely different text style.

Breakout Lab 3.3

Sampling with generate()

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Lab 3.3: Sampling Knobs

Open `day3/lab3_3_sampling/start.ipynb` in the companion repo. Runs offline.

- Generate from one prompt under different settings
- Compare greedy, low and high temperature, top-k, top-p
- Run beam search and note the fluent, safe output
- Match settings to tasks

> Same prompt, different knobs, different text.

Key Takeaways

- [x] Full fine-tuning is expensive in memory and storage per task
- [x] PEFT freezes the base and trains a small adapter
- [x] LoRA learns a low-rank `B x A` update to frozen weights
- [x] QLoRA quantizes the base to 4-bit; it needs a CUDA GPU
- [x] Decoding knobs (temperature, top-k, top-p, beam) shape the output

Questions?

Next: Retrieval-Augmented Generation

Short break, then Session 3

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